

UNIT-III (P.D.E)

PDE: - An equation involving partial diff. coefficient of a function of two or more variables is known as P.D.E

Formation of PDE

1. Elimination of arbitrary constant
2. " " " functions

⑤ Elimination of arbitrary constant

Form a PDE from the following

① $z = ax + by$ — (1)

or diff (1) w.r.t x

$$p = \frac{\partial z}{\partial x} = a$$

$$q = \frac{\partial z}{\partial y} = b$$

$$z = px + qy$$

$$\boxed{z = \frac{\partial z}{\partial x} x + y \frac{\partial z}{\partial y}}$$

② $z = ax + by + ab$ — (1)

$$p = \frac{\partial z}{\partial x} = a$$

$$q = \frac{\partial z}{\partial y} = b$$

Then (1) $\Rightarrow z = px + qy + pq$ (constant)

$$\boxed{z = x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} + \frac{\partial z}{\partial x} \frac{\partial z}{\partial y}}$$

③ $z = (x+a)(y+b)$

$$\text{Here } p = \frac{\partial z}{\partial x} = y+b$$

$$q = \frac{\partial z}{\partial y} = x+a$$

$$\boxed{z = pq}$$

④ $z = \frac{x}{a} + \frac{y}{b}$

$$p = \frac{\partial z}{\partial x} = \frac{1}{a}$$

$$q = \frac{\partial z}{\partial y} = \frac{1}{b}$$

$$\boxed{z = px + qy}$$

⑤ Form a PDE from the relation $z = a \log \left[\frac{b(y-1)}{1-x} \right]$ — (1)

sol Given that

$$z = a \log b + a \log (y-1) - a \log (1-x)$$

Diff w.r.t x partially

$$p = 0 + 0 + a \left(\frac{1}{1-x} \right)$$

$$\boxed{p = \frac{a}{1-x}} \text{ — (2)}$$

Diff w.r.t 'y' partially

$$z = \frac{\partial z}{\partial x}$$

$$= 0 + \frac{a}{y-1} = 0$$

$$\boxed{z = \frac{a}{y-1}} \quad \text{--- (3)}$$

∴ From (3) & (3)

$$(1-x)p = z(y-1)$$

$$\boxed{px + zy = -cp + z}$$

$$\text{①} \rightarrow (1-x)p = z \quad \text{--- (1)}$$

$$\frac{1}{x} \cdot (1-x)p = \frac{z}{x}$$

$$\text{②} \rightarrow 1 \cdot (1-x)p = z$$

$$p \cdot (1-x) = \frac{1}{x} \cdot (1-x)p = \frac{z}{x} = p$$

$$\text{③} \rightarrow p \cdot (1-x)p = z$$

③ & ① more

⑧ Find the PDE of all spheres of radius λ having centres on the xy-plane

$$\boxed{09 = 5}$$

⑧ Equations of the spheres having radius λ whose centres are on xy plane, centres are $(u, v, 0)$

$$(x-u)^2 + (y-v)^2 + (z-0)^2 = \lambda^2$$

Diff ① w.r.t x partially

$$2(x-u) + 0 + 2zp = 0$$

$$\boxed{(x-u) + zp = 0} \quad \text{--- (2)}$$

$$\text{①} \rightarrow \frac{1}{x} \cdot 0 = \frac{1}{x} \cdot \frac{1}{x} = \frac{1}{x^2}$$

$$\left(\frac{1}{x}\right) \cdot \frac{1}{x} = \frac{1}{x^2} \quad \text{--- (e)}$$

$$\text{bly Diff ① w.r.t y partially} \quad \frac{1}{y} \cdot \frac{1}{y} = \frac{1}{y^2} = p$$

$$0 + 2(y-v) + 2zp = 0$$

$$\boxed{(y-v) + zp = 0} \quad \text{--- (3)}$$

$$\frac{1}{y} \cdot \left(\frac{1}{y}\right) \cdot \frac{1}{y} = \frac{1}{y^3} = p$$

$$\left(\frac{1}{y}\right) \cdot \frac{1}{y} = \left(\frac{1}{y}\right) \cdot \frac{1}{y} = \frac{1}{y^2}$$

$$\left(\frac{1}{y}\right) \cdot \frac{1}{y} = \frac{1}{y^2} = p \quad \text{--- (e)}$$

From ②, $(x-u) = -zp$

From ③, $(y-v) = -zp$

Then ① ⇒ $z^2 p^2 + z^2 p^2 + \left[\left(\frac{1}{x}\right)^2 + \left(\frac{1}{y}\right)^2\right] \frac{1}{x^2} = \lambda^2$

$$\boxed{z^2 (p^2 + q^2 + 1) = \lambda^2}$$

$$\left(\frac{1}{x}\right) \cdot \frac{1}{x} = \frac{1}{x^2} = p$$

$$\boxed{p = \frac{1}{x^2}}$$

(4) Elimination of arbitrary function
Form a PDE from the following

① $z = f(x+ay)$ — (1)

$$p = \frac{\partial z}{\partial x} = f'(x) \cdot \frac{\partial x}{\partial x}$$

$$\boxed{p = f'(x) \cdot 1} \text{ — (2)}$$

By $q = \frac{\partial z}{\partial y} = f'(x) \cdot \frac{\partial x}{\partial y} = f'(x) \cdot a$

$$\boxed{q = f'(x) \cdot a} \text{ — (3)}$$

from (2) & (3)

$$q = f'(x) \cdot a$$

$$\boxed{q = pa}$$

(or)

$$\boxed{q = ap}$$

$$\boxed{\frac{\partial z}{\partial y} = a \frac{\partial z}{\partial x}}$$

② $z = f\left(\frac{y}{x}\right)$

$$p = \frac{\partial z}{\partial x} = y \cdot f'\left(\frac{y}{x}\right) \cdot \left(-\frac{y}{x^2}\right)$$

&

$$q = \frac{\partial z}{\partial y} = y f'\left(\frac{y}{x}\right) \cdot \frac{1}{x} + f\left(\frac{y}{x}\right) \cdot 1$$

$$q - f\left(\frac{y}{x}\right) = \frac{y}{x} \cdot f'\left(\frac{y}{x}\right)$$

Since $p = -\frac{y}{x^2} f'\left(\frac{y}{x}\right)$

$$= -\frac{y}{x} \left[\frac{y}{x} f'\left(\frac{y}{x}\right) \right]$$

$$= -\frac{y}{x} \left[q - f\left(\frac{y}{x}\right) \right]$$

$$px = -qy + y f\left(\frac{y}{x}\right)$$

$$\boxed{px + qy = z}$$

③ Form a PDE from $z = f(x^2 + y^2)$

sol Here $p = \frac{\partial z}{\partial x} = f'(x^2 + y^2) (2x)$

$$q = \frac{\partial z}{\partial y} = f'(x^2 + y^2) (2y)$$

$$\frac{p}{2} = \frac{-2}{y}$$

$$\boxed{py + 2x = 0}$$

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④ Eliminate f & g from $z = yf(x) + xg(y)$

sol Given that $yf(x) + xg(y) \text{ --- (1)}$

Here $p = yf'(x) + g(y) \text{ --- (2) } \times x$

$$q = f(x) + xg'(y) \text{ --- (3) } \times y$$

$$\text{--- Here } s = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial p}{\partial y} \text{ or } \frac{\partial q}{\partial x}$$

$$\text{i.e., } s = f'(x) + g'(y) \text{ --- (4)}$$

$$\textcircled{2} \times x + \textcircled{3} \times y$$

$$px + qy$$

$$= xyf'(x) + xg(y) + yf(x) + xyg'(y)$$

$$= xy[f'(x) + g'(y)] + z$$

$$= xy s + z$$

$$\therefore \boxed{z = px + qy - xy s}$$

(P6) Eliminate, $-f(x) + e^y g(x) \rightarrow (1)$

-from (1) $\Rightarrow z = \frac{\partial \tilde{z}}{\partial y} = \phi e^y g(x)$

f

$\tilde{z} = \frac{\partial \tilde{z}}{\partial y} = \frac{\partial z}{\partial y} = e^y g(x)$

$\therefore \boxed{z = \tilde{z}}$

$\frac{\partial z}{\partial y} = \frac{\partial \tilde{z}}{\partial y} \Rightarrow \boxed{\frac{\partial \tilde{z}}{\partial y} - \frac{\partial z}{\partial y} = 0}$

(P6) Form a PDE by eliminating 'f' from the following
Sol $f(xy+z, x+y+z) = 0 \rightarrow (1)$

from (1) $\Rightarrow xy+z = \phi(x+y+z) \rightarrow (2)$

Diff eq (2) w.r.t 'x' partially

$y+2z p = \phi'(x+y+z)(1+p) \rightarrow (3)$

Diff eq (2) w.r.t 'y' partially

$x+2z q = \phi'(x+y+z)(1+q) \rightarrow (4)$

$\frac{\text{eq (3)}}{\text{eq (4)}} \Rightarrow \frac{y+2z p}{x+2z q} = \frac{1+p}{1+q}$

$(y+2z p)(1+q) = (1+p)(x+2z q)$

$y + qy + 2zp + 2zpq = x + 2zq + px + 2pqz$

$\boxed{(x+y) + px - qy + 2z(z-p) = 0}$

NOTE: From all the above Pb we can notice that a PDE can have a sol either in ordinary constant or in ordinary functions

Since, we can form PDE $z = ax + by$, which is same as PDE

$$z = yf\left(\frac{y}{x}\right)$$

Lagrange's Linear equations (Quasi-linear equations)

→ The standard form of this equations is $Pp + Qq = R$

where P, Q, R are functions in x, y, z

→ To solve this eq we write the Auxiliary equations (subsidiary equations) has $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

→ From this we obtained two solutions $u(x, y, z) = C_1$
 $v(x, y, z) = C_2$ by grouping either $\frac{dx}{P} = \frac{dy}{Q}$ (or) $\frac{dy}{Q} = \frac{dz}{R}$ (or) $\frac{dx}{P} = \frac{dz}{R}$

→ Then we write the General sol $\phi(u, v) = 0$

Pb Solve the partial diff equation

$$px + 2y = 3z$$

Sol → It is of the form $px + qy = rz$ (or) It is a quasi-linear eq

→ Now the Auxiliary equation are $\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{3z}$

Taking one & two fraction

$$\frac{dx}{x} = \frac{dy}{y}$$

$$\log x = \log y + \log C_1$$

$$\frac{x}{y} = C_1$$

Taking two & three fractions

$$\frac{dy}{y} = \frac{dz}{3z}$$

$$3 \log y = \log z + \log C_2$$

$$\frac{y^3}{z} = C_2$$

∴ The general solution for the

given eq $\phi\left(\frac{x}{y}, \frac{y^3}{z}\right) = 0$

(Pb) solve the partial diff equation

$$Px + 2y = \sqrt{z}$$

sol It is of the form

$$Pp + Qq = R$$

now Auxiliary equations are

$$\frac{dx}{\sqrt{x}} \pm \frac{dy}{\sqrt{y}} = \frac{dz}{\sqrt{z}}$$

By grouping,

$$\frac{dx}{\sqrt{x}} = \frac{dy}{\sqrt{y}}$$

$$2\sqrt{x} = 2\sqrt{y} + C_1$$

$$\boxed{(\sqrt{x} - \sqrt{y}) = C_1}$$

$$\frac{dy}{\sqrt{y}} = \frac{dz}{\sqrt{z}}$$

$$2\sqrt{y} = 2\sqrt{z} + C_2$$

$$\boxed{(\sqrt{y} - \sqrt{z}) = C_2}$$

$\therefore$ The general sol is $\phi(u, v) = 0$

$$(u) \frac{Pp - Qq}{Q} = \frac{R - P}{P} \text{ i.e., } \phi(\sqrt{x} - \sqrt{y}, \sqrt{y} - \sqrt{z})$$

(Pb)

$$P \tan x + 2 \tan y = \tan z$$

sol It is of the form

$$Pp + Qq = R$$

now Auxiliary equations are

$$\frac{dx}{\tan x} \mp \frac{dy}{\tan y} = \frac{dz}{\tan z}$$

from the first two fractions

$$\frac{dx}{\tan x} = \frac{dy}{\tan y}$$

$$\log |\sin x| = \log |\sin y| + \log C_1$$

$$\log \left[\frac{\sin x}{\sin y} \right] = \log C_1$$

$$\boxed{\frac{\sin x}{\sin y} = C_1}$$

taking 1 & 3 fraction

$$\frac{dx}{\tan x} \mp \frac{dz}{\tan z}$$

$$\log |\sin x| = \log |\sin z| + \log C_2$$

$$\boxed{\frac{\sin x}{\sin z} = C_2}$$

→ The general solution is $\phi(u, v) = 0$

$$\phi\left(\frac{\sin x}{\sin y}, \frac{\sin x}{\sin z}\right) = 0$$

⑩ $x^2 + y^2 = xy$

→ Here Auxiliary equations,

$$\frac{dx}{xz} = \frac{dy}{yz} = \frac{dz}{xy}$$

From first two fractions

$$\frac{dx}{x^2} = \frac{dy}{y^2}$$

$$\frac{dx}{x} = \frac{dy}{y}$$

$$\log x = \log y + \log c_1$$

$$\boxed{\frac{x}{y} = 9} \quad \text{--- (1)}$$

taking ~~two~~ 1, three fraction

$$\frac{dx}{x^2} = \frac{dz}{xy}$$

$$\frac{dx}{x} = \frac{dz}{z}$$

$$y dx - z dz = 0$$

from (i) $\Rightarrow \frac{x}{9} dx - 2dz = 0$

Integrate we get (from 2) $2E$

x

$$\frac{1}{2} \frac{d^2}{dt^2} \left[\frac{1}{2} \frac{d^2}{dt^2} \right] = \frac{1}{2} \frac{d^2}{dt^2} \left[\frac{1}{2} \frac{d^2}{dt^2} \right]$$

$$\frac{x^2}{c_1} - z^2 = c_2$$

$$\boxed{0 = 160C, y^2 - z^2 = C_2} \quad \text{(or)} \quad \boxed{2xy - z^2 = C_2}$$

→ the general sol is $\phi(u,v) = 0$

$$\phi\left(\frac{x}{y}, xy - z^2\right) = 0$$

(P6) $y^2z + x^2z = xy^2$

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Here Auxiliary equations,

$$\frac{dx}{y^2} = \frac{dy}{x^2 z} = \frac{dz}{xy^2}$$

From ① & ③ Fractions

$$\frac{dx}{y^2} = \frac{dz}{2y} \Rightarrow \frac{dx}{2} = \frac{dz}{2}$$

$$x dx = z dz$$

$$\frac{(x^2)}{2} = \frac{y^2}{2} + \frac{C_1}{2} \Rightarrow \boxed{x^2 - y^2 = C_1}$$

$$\frac{dx}{y^2} = \frac{dy}{x^2}$$

$$x^2 dx = y^2 dy$$

$$\frac{x^3}{3} = \frac{y^3}{3} + \frac{C_2}{3}$$

$$\boxed{x^3 - y^3 = C_2}$$

The general solⁿ is $\phi(u, v) = 0$

$$\boxed{\phi(x^3 - y^3, x^3 - y^3) = 0}$$

Method of multipliers

→ For the given equation $Pp + Qq = R$, first we write the A.E.

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R},$$

If (l, m, n) is one set of multipliers can be chosen in

such a way that $\boxed{lp + mq + nR = 0}$ & $l dx + m dy + n dz$ is a

Perfect differential equal to zero

$$\boxed{l dx + m dy + n dz = 0} \quad \text{--- (1)}$$

where $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l dx + m dy + n dz}{lp + mq + nR}$

From (1) $\boxed{u(x, y, z) = C_1}$ (one solⁿ)

Similarly we identify another set of multipliers (l', m', n')

$$\Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l' dx + m' dy + n' dz}{l'P + m'Q + n'R}$$

$$\& \quad l'P + m'Q + n'R = 0,$$

→ then $l' dx + m' dy + n' dz = 0$

we get another solution,

$$\boxed{v(x, y, z) = C_2}$$

∴ general solⁿ is $\phi(u, v) = 0$

(Pb) solve the P.D.E $(y+z)^p - xyz + xz = 0$

Sol Given eq can be written as

$$(y+z)^p - xyz = -xz \quad \text{--- (1)}$$

Auxiliary equations are

$$\frac{dx}{(y+z)^p} = \frac{dy}{(-xy)} = \frac{dz}{(-xz)} \quad \text{--- (2)}$$

from (2) & (3) fractions

$$\frac{dy}{-xy} = \frac{dz}{-xz} \Rightarrow \frac{dy}{y} = \frac{dz}{z}$$

$$\log y = \log z + \log C_1$$

$$\boxed{\frac{y}{z} = C_1} \quad (\text{one sol})$$

We choose (x, y, z) as multipliers then each fraction into in (2) is equal to

$$\frac{x dx + y dy + z dz}{xy^2 + xz^2 - xy^2 - xz^2}$$

$$\frac{dx}{y+z^p} = \frac{dy}{-xy} = \frac{dz}{-xz} = \frac{x dx + y dy + z dz}{0} \quad \text{--- (3)}$$

$$x dx + y dy + z dz = 0$$

Integrate we get

$$\boxed{x^2 + y^2 + z^2 = C_2}$$

$\therefore$ The general solution is

$$\boxed{\phi\left(\frac{y}{z}, x^2 + y^2 + z^2\right) = 0}$$

(Pb) solve the partial diff eq $zp + yq = x$ --- (1)

Sol Auxiliary equations are,

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z} \quad \text{--- (2)}$$

from (1) & (2) fractions

$$\frac{dx}{x} = \frac{dz}{z}$$

$$x dx = z dz$$

$$\frac{x^2}{2} = \frac{z^2}{2} + \frac{C_1}{2} \Rightarrow \boxed{x^2 - y^2 = C_1}$$

We use the multipliers (x, y, z) , then each fraction in (1) is equal to

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z} = \frac{dx+dy+dz}{x+y+z}$$

$$\frac{dy}{y} = \frac{d(x+y+z)}{x+y+z}$$

Integrating we get

$$\log y = \log(x+y+z) + \log C_2$$

$$\frac{y}{x+y+z} = C_2$$

The general sol is are

$$\phi\left(x^2 - z^2, \frac{y}{x+y+z}\right) = 0$$

(P6) Solve the PDE $(x+y)z^2 p + (x-y)z^2 q = x^2 + y^2$ — (1)

Auxiliary equation is

$$\frac{dx}{(x+y)z} = \frac{dy}{(x-y)z} = \frac{dz}{x^2+y^2} \quad \text{--- (2)}$$

using the multipliers (x, y, z) then each fraction in (2) is

$$\begin{aligned} & \frac{xdx + ydy + zdz}{x^2 + xy + y^2 + x^2 + xy + y^2 - \cancel{x^2 + xy + y^2} + x^2 + y^2} \\ &= \frac{xdx + ydy + zdz}{2x^2 + 2y^2} \end{aligned}$$

$$\Rightarrow \frac{dx}{(x+y)z} = \frac{xdx + ydy + zdz}{2x^2 + 2y^2}$$

$$xdx - ydy - zdz = 0$$

Integrating we get

$$\frac{x^2}{2} - \frac{y^2}{2} - \frac{z^2}{2} = C_1 \quad \text{--- (3)}$$

we use another multiplier $(y, z, -2)$ then each fraction in

② is equal to

$$\frac{ydx + xdy - 2dz}{xy^2 + y^2z + z^2 - zy^2 - x^2z - y^2z}$$

$$ydx + xdy - 2dz = 0$$

$$\boxed{d(xy) - 2dz = 0}$$

Integrating we get

$$\boxed{xy - \frac{z^2}{2} = C}$$

general sol is

$$\boxed{\phi\left(x^2 - y^2 - z^2, xy - \frac{z^2}{2}\right) = 0}$$

Multipliers

① $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$

② $x(y - z)p + y(z - x)q = z(x - y)$

Non Homogeneous Partial Diff equation

(Homogeneous linear partial Diff equation) with constant coefficients.

→ The standard form of this equation is

$$a_0 \frac{\partial^2 z}{\partial x^2} + a_1 \frac{\partial^2 z}{\partial x \partial y} + a_2 \frac{\partial^2 z}{\partial y^2} + \dots + a_n \frac{\partial^2 z}{\partial y^n} = f(x, y) \quad (1)$$

→ Taking $D = \frac{\partial}{\partial x}$, $D' = \frac{\partial}{\partial y}$, $D^3 = \frac{\partial^3}{\partial x^3}$

$$D' = \frac{\partial}{\partial y}, \quad D'^2 = \frac{\partial^2}{\partial y^2}, \quad D'^3 = \frac{\partial^3}{\partial y^3}$$

we have $(a_0 D^n + a_1 D^{n-1} D' + a_2 D^{n-2} D'^2 + \dots + a_n D^n) z = f(x, y) \quad (2)$

(or)

$$\boxed{f(D, D') z = f(x, y)}$$

→ Auxiliary equation: Dividing D^n on both sides $f(D, D') = 0$

we get, $\boxed{a_0 \left(\frac{D}{D'}\right)^n + a_1 \left(\frac{D}{D'}\right)^{n-1} + a_2 \left(\frac{D}{D'}\right)^{n-2} + \dots + a_n = 0}$

∴ which is the auxiliary equation

Let $\frac{D}{D'} = m$

so that $\boxed{a_0 m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_n = 0}$

roots be $m_1, m_2, \dots, m_n$

Case (i): If all roots are real & distinct then

$$C.F = \phi_1(y + m_1 x) + \phi_2(y + m_2 x) + \dots + \phi_n(y + m_n x)$$

Case (ii): If all roots are equal $m_1 = m_2 = \dots$

$$C.F = \phi_1(y + m_1 x) + x \phi_2(y + m_1 x) + x^2 \phi_3(y + m_1 x) + \dots$$

we

∴ Here CF is the solution of $f(D, D')Z = 0$

Particular Integral :- The solution of $f(D, D')Z = F(x, y)$ w.r.t

inverse operator form $\frac{1}{f(D, D')}$

$$\text{i.e., } \boxed{Z = \frac{1}{f(D, D')} F(x, y)}$$

∴ The General Sol of eq(1)

$$\boxed{Z = CF + P.I}$$

Problems on Finding

(P6) Solve the PDE $\frac{\partial^2 z}{\partial x^2} + 4 \frac{\partial^2 z}{\partial x \partial y} - 5 \frac{\partial^2 z}{\partial y^2} = 0$

Sol Taking $D = \frac{\partial}{\partial x}$ $D' = \frac{\partial}{\partial y}$

So that given equation will be

$$(D^2 + 4DD' - 5D'^2)Z = 0$$

Here $f(D, D') = D^2 + 4DD' - 5D'^2$

Auxiliary equation $f(m, 1) = 0$

$$m^2 + 4m - 5 = 0$$

$$m = 1, -5$$

$$\therefore Z = \phi_1(y+x) + \phi_2(y-5x)$$

(P6) Solve the diff equation $2 \frac{\partial^2 z}{\partial x^2} + 5 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = 0$

Sol Taking $D = \frac{\partial}{\partial x}$ $D' = \frac{\partial}{\partial y}$

So that given equation will be

$$(2D^2 + 5DD' + 2D'^2)Z = 0$$

Here $f(D, D') = 2D^2 + 5DD' - 2D'^2$

Auxiliary equation is

$$f(m, 1) = 0$$

i.e., $2m^2 + 5m + 2 = 0$

$$\therefore \begin{cases} D = m \\ D = 1 \end{cases}$$

$$m = -2, -\frac{1}{2}$$

C.F

$$\boxed{z = \phi_1(y - 2x) + \phi_2(y - \frac{x}{2})}$$

(Pb) $(D^3 - 4DD' + 3D'D'')z = 0$

sol $f(D, D') = D^3 - 4DD' + 3D'D''$

Auxiliary equation are

$$D^3 - 4DD' + 3D'D'' = 0$$

$$f(m, 1) = 0$$

i.e., $m^3 - 4m + 3 = 0$

$$m[m^2 - 4m + 3] = 0$$

$$m = 0, 1, 3$$

C.F

$$\boxed{z = \phi_1(y + 0x) + \phi_2(y + x) + \phi_3(y + 3x)}$$

(Pb) Solve the PDE $\frac{\partial^4 z}{\partial x^4} + \frac{\partial^4 z}{\partial y^4} = 2 \frac{\partial^4 z}{\partial x^2 \partial y^2}$

Given eq in operator form

$$(D^4 - 2D^2D' + D'^4)z = 0$$

$$f(D, D') = D^4 - 2D^2D' + D'^4$$

Auxiliary equation are

$$f(m, 1) = 0$$

i.e., $m^4 - 2m^2 + 1 = 0$

$$(m^2 - 1)^2 = 0$$

$$m = \pm 1, \pm 1$$

C.F

$$z = \phi_1(y+x) + x\phi_2(y+x) + \phi_3(y-x) + x\phi_4(y-x)$$

$$\textcircled{c} \frac{\partial^4 z}{\partial x^4} - \frac{\partial^4 z}{\partial y^4} = 0$$

$$0 = m^4 - (-m)^4 = 0$$

Sol. Operator form: $(D^4 - D'^4)z = 0$

Auxiliary equation: $D^4 - D'^4 = 0$

$$(x^4 - 1) = 0$$

i.e., $m^4 - 1 = 0$

$$(m^2 - 1)(m^2 + 1) = 0$$

$$m = \pm 1, \pm i$$

C.F

$$z = \phi_1(y+x) + \phi_2(y-x) + \phi_3(y+ix) + \phi_4(y-ix)$$

$$\frac{(p^2 - x^2)z}{x} + (p^2 - x^2)z + (x^2 - p^2)z + (p^2 - x^2)z = 0$$

7.0

Q6 $(D^3 - 4DD' + 4DD'^2)z = \sin(3x+2y)$

Auxiliary equation: $f(m, l) = 0$

$$m^3 - 4ml + 4ml^2 = 0$$

$$0 = \frac{\delta'''}{\delta''} - \frac{\delta''}{\delta} \quad (1)$$

$$m(m^2 - 4m + 4) = 0$$

$$m = 0, 2, 2$$

$$C.F = \phi_1(y) + \phi_2(y+2x) + x\phi_3(y+2x)$$

$$P.F = \frac{1}{D^3 - 4DD' + 4DD'^2} \cdot \sin(3x+2y)$$

$$= \frac{1}{D(-2^3) + 4D(-6) + 4D(-2^2)} \cdot \sin(3x+2y)$$

replace

$$\begin{aligned} D &\rightarrow -a \\ DD' &\rightarrow -ab \\ D'^2 &\rightarrow -b^2 \end{aligned}$$

$$= \frac{1}{D[-9+24-16]} \cdot \sin(3x+2y)$$

$$= -\frac{1}{D} \sin(3x+2y) = \frac{\cos(3x+2y)}{3}$$

$$C.S = \phi_1(y) + \phi_2(y+2x) + x\phi_3(y+2x) + \frac{\cos(3x+2y)}{3}$$

Q6 $(D'' - DD')z = \sin(x)\cos(2y)$

Auxiliary equation: $D'' - DD' = 0$

$$m'' - m = 0$$

$$m(m-1) = 0$$

$$m = 0, 1$$

$$C.F = \phi_1(y) + \phi_2(y+x)$$

$$P.I = \frac{1}{D^2 - DD'} \cdot \sin x \cdot \cos 2y$$

replace $\Rightarrow D^2 \rightarrow -a^2$

multiply & divide by '2'

$+DD' \rightarrow -ab$

$$= \frac{2}{2} \left[\frac{1}{D^2 - DD'} \cdot \sin x \cdot \cos 2y \right]$$

$$= \frac{1}{2} \left[\frac{1}{D^2 - DD'} \sin(x+2y) + \frac{1}{D^2 - DD'} \sin(x-2y) \right]$$

$$= \frac{1}{2} \left[\frac{1}{-1+2} \sin(x+2y) + \frac{1}{-1+2} \sin(x-2y) \right]$$

$$P.I = \frac{1}{2} \left[\frac{\sin(x+2y)}{1} + \frac{\sin(x-2y)}{-3} \right]$$

$$C.F = \phi_1(y) + \phi_2(y+x) + \frac{1}{2} \left[\sin(x+2y) - \frac{\sin(x-2y)}{3} \right]$$

$$\textcircled{Pb} (D^3 - uD^2D' + uDD'^2)z = u \sin(2x+y)$$

Sol Auxiliary equation: $f(m, n) = 0$

$$m^3 - um^2 + um = 0$$

$$\boxed{m = 0, 2, 2}$$

~~A.E~~ P.S

$$C.F = \phi_1(y) + \phi_2(y+2x) + x \phi_3(y+2x)$$

$$P.I = 4 \cdot \frac{1}{D^3 - uD^2D' + uDD'^2} \cdot \sin(2x+y)$$

$$= 4 \cdot \frac{1}{D^3 - uD^2D' + uDD'^2} \sin(2x+y)$$

(Failure case)

$$P.I = 4 \cdot \frac{1}{D^3 - 8DD'^2 + 4DD'^2} \sin(2x+y)$$

$$= x \cdot \frac{1}{3(-2^2) + (-8)(-2) + 4(-1^2)} \sin(2x+y)$$

$$= x \cdot \frac{1}{-12+16-4} \sin(2x+y)$$

(failure case)

$$\therefore P.T = 4x^2 \cdot \frac{1}{60-80^2} \sin(2x+y)$$

$$= 2x^2 \frac{(80+40^2)}{90^2-160^2} \sin(2x+y)$$

$$= 8x^2 \frac{(80+40^2)}{9(-2^2)-16(-1^2)} \sin(2x+y)$$

$$= 2x^2 \left[\frac{6 \cos(2x+y) + 4 \cos(2x+y)}{-20} \right]$$

$$= \frac{-x^2 [6 \cos(2x+y) + 4 \cos(2x+y)]}{10}$$

$$G.S = \phi_1(y) + \phi_2(y+2x) + x(y+2x) + \frac{x^2}{2} [6 \cos(2x+y) + 4 \cos(2x+y)]$$

$$(Pb) (D'' + DD' - 6D'^2) Z = x+y$$

$$A.E: m^2 + m - 6 = 0$$

$$\begin{matrix} -6 \\ 2 \quad 3 \end{matrix}$$

$$m \neq x+y$$

$$m = -3, 2$$

$$G.F = \phi_1(y+2x) + \phi_2(y-3x)$$

$$P.T = \frac{1}{D'' + DD' - 6D'^2} \cdot (x+y)$$

$$= \frac{1}{D''} \left[1 + \left(\frac{DD' - 6D'^2}{D''} \right) \right]^{-1} (x+y)$$

$$= \frac{1}{D''} \left[1 - \left(\frac{DD' - 6D'^2}{D''} \right) + \dots \right]$$

$$= \frac{1}{D''} \left[(x+y) - \frac{D'}{D} (x+y) - \dots \right]$$

$$= \frac{1}{D''} \left[(x+y) - x - 0 \right]$$

$$= \frac{1}{D''} [y] = y \int dx \, dx$$

$$G.S = \phi_1(y+2x) + \phi_2(y-3x)$$

$$(Pb) (D'' + 2DD' + D'^2) Z = x^2 + xy + y^2$$

$$Sol \, A.E: f(m,1) = 0$$

$$m^2 + 2m + 1 = 0$$

$$m = -1, -1$$

$$G.F = \phi_1(y-x) + x \phi_2(y-x)$$

$$P.T = \frac{1}{D'' + 2DD' + D'^2} \cdot (x^2 + xy + y^2)$$

$$= \frac{1}{D''} \left[1 + \left(\frac{2DD' + D'^2}{D''} \right) \right]$$

$$= \frac{1}{D''} \left[1 - \left(\frac{2DD' + D'^2}{D''} \right) \right]$$

$$P.I = \frac{1}{D^2 + DD' - 6D'^2} \cdot (x+y)$$

$$= \frac{1}{D^2} \left[1 + \frac{(DD' - 6D'^2)}{D^2} \right]^{-1} (x+y)$$

$$= \frac{1}{D^2} \left[1 - \left(\frac{DD' - 6D'^2}{D^2} \right) + \dots \right] (x+y)$$

$$= \frac{1}{D^2} \left[(x+y) - \frac{D'}{D} (x+y) - \left(\frac{D'}{D} \right)^2 (x+y) \right]$$

$$= \frac{1}{D^2} \left[(x+y) - x - 0 \right]$$

$$= \frac{1}{D^2} [y] = y \int dx \, dx = y \int x \, dx = \frac{yx^2}{2}$$

$$\boxed{C.S = \phi_1(y+2x) + \phi_2(y-3x) + \frac{yx^2}{2}}$$

Q6) $(D^2 + 2DD' + D'^2)z = x^2 + xy + y^2$

Sol A.E: $f(m,1) = 0$

$$m^2 + 2m + 1 = 0$$

$$m = -1, -1$$

$$\boxed{C.F = \phi_1(y-x) + x\phi_2(y-x)}$$

$$P.I = \frac{1}{D^2 + 2DD' + D'^2} \cdot (x^2 + xy + y^2)$$

$$= \frac{1}{D^2} \left[1 + \left(\frac{2DD' + D'^2}{D^2} \right) \right]^{-1} (x^2 + xy + y^2)$$

$$= \frac{1}{D^2} \left[1 - \left(\frac{2DD' + D'^2}{D^2} \right) + \left(\frac{2DD' + D'^2}{D^2} \right)^2 + \dots \right] (x^2 + xy + y^2)$$

$$= \frac{1}{D^2} \left[1 - 2\frac{D'}{D} - 2\left(\frac{D'}{D}\right)^2 + 3\left(\frac{D'}{D}\right)^3 \right] (x^2 + xy + y^2)$$

$$= \frac{1}{D^2} \left[(x^2 + xy + y^2) - 2\frac{D'}{D} (x^2 + xy + y^2) + 3\frac{D'^2}{D^2} (x^2 + xy + y^2) \right]$$

$$= \frac{1}{D^2} \left[(x^2 + xy + y^2) - \frac{2}{D} (x + 2y) + \frac{3}{D^2} \right]$$

$$= \frac{1}{D^2} \left[(x^2 + xy + y^2) - 2\left(\frac{x^2}{2} + 2yx\right) + \frac{3}{2}\left(\frac{x^2}{2}\right) \right]$$

$$= \frac{1}{D^2} \left[(x^2 + xy + y^2) - x^2 - 4xy + 3x^2 \right]$$

$$= \frac{1}{D} \cdot \frac{1}{D} \left[(x^2 + xy + y^2) - 4xy + 2x^2 \right]$$

$$= \frac{1}{D} \cdot \left[\frac{x^3}{3} + y\frac{x^2}{2} + 6 - 2yx^2 + \frac{2x^3}{3} \right]$$

$$= \frac{x^4}{12} + \frac{x^2 y^2}{2} - \frac{4x^2}{2}$$

$$= \frac{x^2 y^2}{2} + \frac{x^4}{12} - 2x^2$$

$$C_{1.8} = \phi_1(y-x) + x\phi_2(y-x) + \left[\frac{x^2 y^2}{2} + \frac{x^4}{12} - 2x^2 \right]$$

Case (iv): General method

$$(D^2 - 2DD' + D'^2)z = \tan(x+y)$$

Sol

$$A.E : f(m,1) = 0$$

$$m^2 - 2m + 1 = 0$$

$$m = 1, 1$$

$$C.F = \phi_1(y+x) + x\phi_2(y+x)$$

$$P.I = \frac{1}{(D-D')^2} \cdot \tan(x+y)$$

$$\begin{cases} y+x=c \\ y=c-x \end{cases}$$

$$= \frac{1}{(D-D')} \left[\frac{1}{D-D'} (\tan(x+c-y)) \right]$$

$$= \frac{1}{D-D'} \left[\frac{1}{D-D'} (\tan(c)) \right]$$

$$= \frac{1}{D-D'} \cdot \int \tan c \, dx = \int x \tan c \, dx = \int x \cdot \tan(c+y+x) \, dx$$

$$= \int x \cdot \tan(c) \, dx$$

$$= \frac{x^2}{2} \tan c = \frac{x^2}{2} \tan(c+y+x)$$

$$\boxed{C.S = \phi_1(y+x) + x\phi_2(y+x) + \frac{x^2}{2} \tan(c+y+x)}$$

$$(Pb) (D^2 - DD' + 2D'^2)Z = (y-1)e^x$$

$$\therefore A.E \stackrel{P}{=} f(m,1) = 0$$

$$m^2 - m - 2 = 0$$

$$m = -1, 2$$

$$\boxed{C.F = \phi_1(y-x) + \phi_2(y+2x)}$$

$$P.I = \frac{1}{D^2 - DD' - 2D'^2} \cdot (y-1)e^x$$

$$= \frac{1}{(D+D')(D-2D')} (y-1)e^x$$

$$= \frac{1}{D+D'} \left[\frac{1}{D-2D'} (y-1)e^x \right]$$

$$\therefore \begin{cases} y+2x=c \\ y=c-2x \end{cases}$$

$$= \frac{1}{D+D'} \left[\int e^x (c-2x-1) \, dx \right]$$

$$= \frac{1}{D+D'} [ce^x - 2e^x(x-1) - e^x]$$

$$= \frac{1}{D+D'} [(y+2x)e^x - 2e^x(x-1) - e^x]$$

$$= \frac{1}{D+D'} [(y+2x)e^x - 2xe^x + 2e^x - e^x]$$

$$= \frac{1}{D+D'} [ye^x + e^x]$$

$$\therefore \begin{cases} y-x=c \\ y=c+x \end{cases}$$

$$= \int (e^x(c+x) + e^x) dx$$

$$= \int e^x [c+x+1] dx$$

$$= ce^x + e^x(x+1) + e^x$$

$$= ce^x + xe^x - \cancel{e^x} + e^x$$

$$= (y-x)e^x + xe^x = ye^x - \cancel{xe^x} + xe^x$$

$$= ye^x$$

$$\boxed{C_1 = \phi_1(y-x) + \phi_2(y+2x) + ye^x}$$

(Pb) $(D^2 - 5D + 2)\vec{z} = 2y(y-x)$

(Pb) $(D^2 + D - 6D^2)\vec{z} = y \cos x$

A.E : $f(m,1) = 0$

$$2m^2 - 5m + 2 = 0$$

$$m = 2, \frac{1}{2}$$

$$\boxed{C.F = \phi_1(y+2x) + \phi_2(y+\frac{x}{2})}$$

$$P.T = \frac{24}{2D'' - 5DD' + 2D'^2} \cdot (y-x)$$

$$= \frac{24}{2D''} \left[1 - \left[\frac{5DD' + 2D'^2}{2D''} \right] \right]^{-1} (y-x)$$

$$= \frac{24}{2D''} \left[1 + \left(\frac{5DD' - 2D'^2}{2D''} \right) \right] (y-x)$$

$$= \frac{12}{D''} \left[1 + \frac{5}{2} \left(\frac{D'}{D} \right) - \left(\frac{D'}{D} \right)^2 \right] (y-x)$$

$$= \frac{12}{D''} \left[1 + \frac{5}{2} \frac{y}{x} - \frac{x^2}{2} \right] (y-x)$$

$$= \frac{12}{D''} \left[(y-x) + \frac{5}{2} (xy - x^2) \right]$$

$$= \frac{12}{D''} \left[yx + \frac{5}{2} (1) \right]$$

$$= \frac{12}{D''} \left[y + \frac{5x}{2} \right] = \frac{12}{D} \left[xy + \frac{3x^2}{4} \right] = 12 \left[\frac{yx^2}{2} + \frac{x^3}{4} \right]$$

$$= 6x^2y + 3x^3$$

$$C.S = \phi_1(y+2x) + \phi_2\left(y+\frac{x^2}{2}\right) + 6x^2y + 3x^3$$

$$(Pb) (D'' + DD' - 6D'^2)z = y \cos x$$

Sol A.E : $f(m) = 0$

$$m'' + m - 6 = 0$$

$$m = 2, -3$$

-6
↑
32

$$C.F = \phi_1(y+2x) + \phi_2(y-3x)$$

$$P.F = \frac{1}{D^2 + DD' - 6D^2} \cdot y \cos x$$

$$= \frac{1}{(D-2D')(D+3D')} y \cdot \cos x$$

$$= \frac{1}{(D-2D')} \left[\frac{1}{D+3D'} y \cdot \cos x \right]$$

$$= \frac{1}{D-2D'} \left[\int (C+3x) \cos x \, dx \right] \quad \because \begin{cases} y-3x=C \\ y=C+3x \end{cases}$$

$$= \frac{1}{D-2D'} \left[C \sin x + 3(x \sin x + \cos x) \right]$$

$$= \frac{1}{D-2D'} \left[(y-3x) \sin x + 3x \sin x + 3 \cos x \right]$$

$$= \frac{1}{D-2D'} \left[y \sin x + 3 \cos x \right]$$

$$= \int [(C-2x) \sin x + 3 \cos x] \, dx$$

$$= \int (C \sin x - 2x \sin x + 3 \cos x) \, dx \quad \because \begin{cases} y+2x=C \\ y=C-2x \end{cases}$$

$$= -C \cos x - 2(-x \cos x + \sin x) + 3 \sin x$$

$$= -(y+2x) \cos x + 2x \cos x - 2 \sin x + 3 \sin x$$

$$= -y \cos x - 2x \cos x + 2x \cos x - 2 \sin x + 3 \sin x$$

$$= \sin x - y \cos x$$

$$\boxed{G.P = \phi_1(y+2x) + \phi_2(y-3x) + \sin x - y \cos x}$$

Non-Homogeneous partial diff equation

Eg of the form $f(D, D')Z = F(x, y)$ where $f(D, D')$ is not homogeneous is called Non-Homogeneous linear PDE

For this equations the complete solution is C.F + P.I

$$\boxed{G.S = C.F + P.I}$$

The CF procedure is diff from homogeneous equation. The which are Homogeneous linear PDE but P.I procedure is same

To find CF we factorize $f(D, D')$ into the factors of the form

$$D - mD' - c$$

consider, then $\boxed{Z = e^{cx} \phi(y + mx)}$; $(D - mD' - c)Z = 0$

(Pb) Solve the PDE

$$(D^2 + DD' + D' - 1)Z = e^{-x}$$

Sol. here $f(D, D') = D^2 + DD' + D' - 1$

$$= (D-1)(D+1) + D'(D+1)$$

$$= (D+1)(D+D'-1)$$

Now $(D+1)Z = 0$

$$\boxed{c = -1, m = 0}$$

$$Z = e^{-x} \phi_1(y)$$

Now $(D+D'-1)Z = 0$

$$\boxed{m = -1, c = 1}$$

$$Z = e^{-x} \phi_2(y-x)$$

Thus C.F = $e^{-x} \phi_1(y) + e^{-x} \phi_2(y-x)$

Now P.I = $\frac{1}{D^2 + DD' + D' - 1} e^{-x + 0y}$

$$\boxed{a = -1, b = 0}$$

$$= \frac{1}{(-1)^2 + 1(-1)(0) - 1} \cdot e^{-x}$$

(failure)

$$P.I = x \cdot \frac{1}{2D+D'} e^{-x}$$

$$= \frac{-xe^{-x}}{2}$$

$$G.S = e^{-x} \phi_1(y) + e^x \phi_2(y-x) = \frac{xe^{-x}}{2}$$

$$* \textcircled{Pb} (D^2 - 3D - D' + 3D') \bar{z} = e^{x-2y}$$

$$\text{Sol} \quad \text{Here } D^2 - 3D - D' + 3D' = (D-D')(D+D') - 3(D-D') \\ = (D-D')(D+D'-3)$$

$$\text{Now } D-D' = 0$$

$$\bar{z} = e^{0x} \phi_1(y+x) \quad \begin{matrix} c=0 \\ m=-1 \end{matrix}$$

$$\text{Now } D+D'-3=0$$

$$\bar{z} = e^{3x} \phi_2(y-x) \quad \begin{matrix} c=3 \\ m=-1 \end{matrix}$$

$$C.F = e^{0x} \phi_1(y+x) + e^{3x} \phi_2(y-x)$$

$$P.I = \frac{1}{D^2 - 3D - D' + 3D'} \cdot e^{x-2y}$$

$$= \frac{1}{1 - 3(1) - (-2) + 3(-2)} \cdot e^{x-2y} \quad \begin{matrix} a=1 \\ b=-2 \end{matrix}$$

$$= \frac{1}{1+3-4-6} \cdot e^{x-2y}$$

$$= \frac{e^{x-2y}}{-12}$$

$$\boxed{G.S = e^{0x} \phi_1(y+x) + e^{3x} \phi_2(y-x) - \frac{e^{x-2y}}{12}}$$

$$(b) (D^2 - DD' - 2D)z = \sin(3x+4y)$$

$$\text{or here } D^2 - DD' - 2D = D[D - D' - 2]$$

$$C.F = e^{ax} \phi_1(y) + e^{bx} \phi_2(y+x)$$

$$P.I = \frac{1}{D^2 - DD' - 2D} \cdot \sin(3x+4y)$$

$$= \frac{1}{D^2 - D' - (-2) - 2D} \cdot \sin(3x+4y) \quad \begin{matrix} D^2 = -9 \\ -DD' = 12 \\ +6^2 = -4 \end{matrix}$$

$$= \frac{1}{-9+12-2D} \cdot \sin(3x+4y)$$

$$= \frac{1}{3-2D} \sin(3x+4y)$$

$$= \frac{1}{3-2D} \times \left(\frac{3+2D}{3+2D} \right) \cdot \sin(3x+4y)$$

$$= \frac{3+2D}{9-4D^2} \cdot \sin(3x+4y)$$

$$= \frac{3+2D}{9-4(-9)} \cdot \sin(3x+4y)$$

$$= \frac{3+2D}{9(5)} \cdot \sin(3x+4y)$$

$$= \frac{3+2D}{45} \cdot \sin(3x+4y)$$